Lesson plan · for the teacher

Code on paper

Programming · Level 4 · Lesson 3

Time
40 min

Materials
Printed worksheets or the board; Paper strips and scissors; One phone to check the answers

Learning goals

- Use Parsons strips, trace tables and desk checks.
- Assess programming fairly on paper.
- Use Darsi's printed worksheets.

Activity: Paper first

1. Warm-up (5 min): Write the while loop that adds 1 to 3 on the board. Pairs write its output on paper.
2. Trace (10 min): Pupils draw a table with columns n and total, and fill one row per turn. Check the first two rows together.
3. Strips (10 min): Pairs order and indent a set of Parsons strips, with one distractor to reject.
4. Desk check (10 min): Pairs swap their trace tables and strips and check each other's, circling any disagreement.
5. Wrap-up (5 min): Run the programs on one phone and settle the disagreements. Then give the exit ticket.

This lesson works with no power and no devices; the phone at the end is optional.

Discussion

- What did the trace tables show that running the code wouldn't?
- How could you assess your next unit on paper?
- Which pupils benefit most from paper-first lessons?

Exit ticket

- Write one paper question that tests loops, and its answer.

Try it in class tomorrow
Print (or copy onto the board) this lesson's worksheet. Pairs do the trace table and the strips on paper; at the end, one phone checks the answers.

Worksheet · for pupils

Code on paper

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Name _____ Class _____ Date _____

1 **Half your class has no phone at home. Which homework is fairest?**

- ◯ Type and run a program at home
- ◯ A printed trace table and Parsons strips
- ◯ Watch a video online

2 **Paper strips for a class: the program adds 1, 2 and 3 and prints Total: 6. One strip doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

- ☐ `print("Total:", total)`
- ☐ `total = total + n`
- ☐ `for n in range(1, 3):`
- ☐ `total = 0`
- ☐ `for n in range(1, 4):`

3 **A row of the trace table. After the loop has run twice, what are n and total?**

```
n = 1
total = 0
while n <= 3:
    total = total + n
    n = n + 1
```

◯ n = 3, total = 3 ◯ n = 2, total = 3 ◯ n = 3, total = 6 ◯ n = 2, total = 2

4 **A gap-fill for paper: the program counts the marks of 10 or more.**

```
marks = [12, 7, 15]
passed = 0
for m in marks:
    if m _____ 10:
        passed = _____
print(passed)
```

Word bank: `m + 1` `=` `passed + 1` `>=`

5 **A desk check: this loop should add 1 to 3, but it never stops. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  n = 1
2  total = 0
3  while n <= 3:
4      total = total + n
5      n = n - 1
6  print(total)
```

✎ **Which part of your next programming unit could be done entirely on paper? Which part really needs a device, and why?**
